

NAME

sched_yield – yield the processor

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sched.h>
```

```
int sched_yield(void);
```

DESCRIPTION

sched_yield() causes the calling thread to relinquish the CPU. The thread is moved to the end of the queue for its static priority and a new thread gets to run.

RETURN VALUE

On success, **sched_yield()** returns 0. On error, *-1* is returned, and *errno* is set to indicate the error.

ERRORS

In the Linux implementation, **sched_yield()** always succeeds.

STANDARDS

POSIX.1-2008.

HISTORY

POSIX.1-2001 (but optional). POSIX.1-2008.

Before POSIX.1-2008, systems on which **sched_yield()** is available defined **_POSIX_PRIORITY_SCHEDULING** in *<unistd.h>*.

CAVEATS

sched_yield() is intended for use with real-time scheduling policies (i.e., **SCHED_FIFO** or **SCHED_RR**). Use of **sched_yield()** with nondeterministic scheduling policies such as **SCHED_OTHER** is unspecified and very likely means your application design is broken.

If the calling thread is the only thread in the highest priority list at that time, it will continue to run after a call to **sched_yield()**.

Avoid calling **sched_yield()** unnecessarily or inappropriately (e.g., when resources needed by other schedulable threads are still held by the caller), since doing so will result in unnecessary context switches, which will degrade system performance.

SEE ALSO

sched(7)